Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 30 epochs with a batch size of 32. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the f1 score of MLP by dataset.

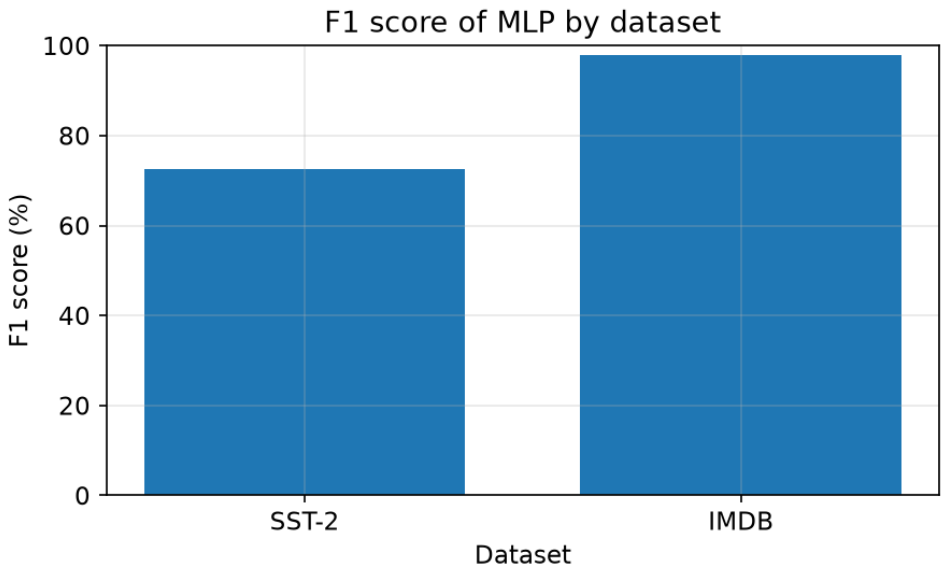


Figure 1: F1 score of MLP by dataset for the models in this experiment.

MLP scores 45.2 points higher on IMDB than on SST-2. On SST-2, MLP obtains a F1 score of over 70% (Figure 1).

Figure 2 shows the validation accuracy during training.

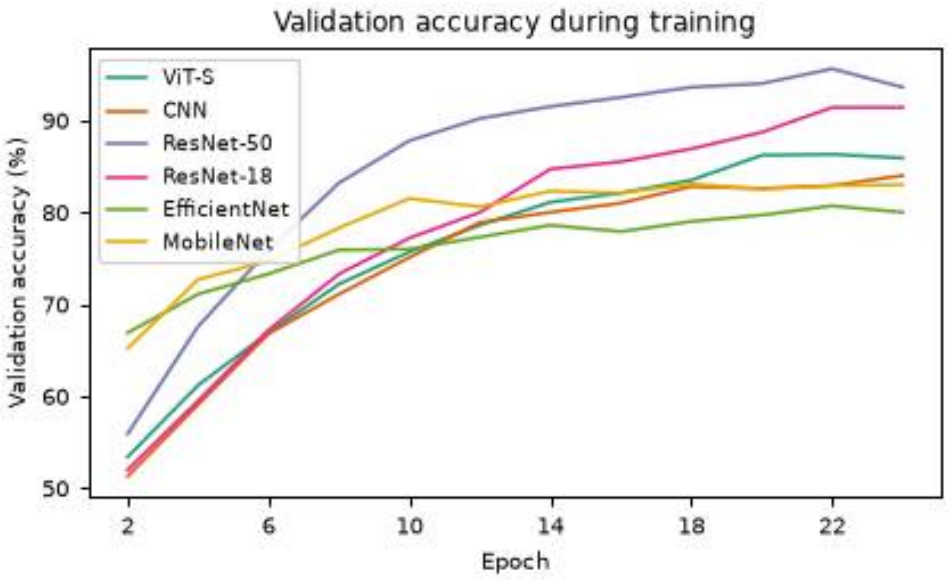


Figure 2: Validation accuracy during training for the models in this experiment.

At epoch 8, ResNet-50 achieves 83.2% validation accuracy. ResNet-18 attains a validation accuracy of about 70% at epoch 8. At epoch 12, EfficientNet achieves 77.3% validation accuracy (Figure 2). ResNet-50 attains a validation accuracy of 91.5% at epoch 14 (Figure 2).

Figure 3 shows the validation accuracy during training.

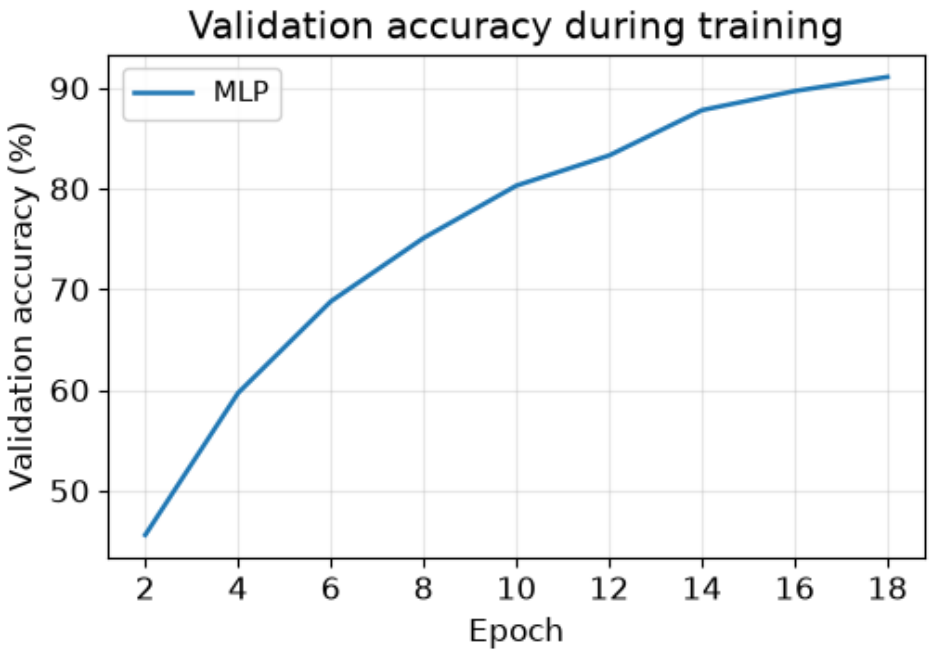


Figure 3: Validation accuracy during training for the models in this experiment.

As shown in Figure 3, MLP scores 24.6 points higher on 16 than on 8. MLP attains a validation accuracy of 83.3% at epoch 12. MLP attains a validation accuracy of 45.6% at epoch 2. MLP scores 24 points higher on 12 than on 4.

3 Discussion

On QQP, Random Forest obtains an accuracy of 50.4%. We note that the training was stable for all models.